Project Proposal

Project Title: Community Scam Pattern Analysis and Investigation Database
Group: Group 10

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1. Project Aims

Our main goal is to build a central system where everyday people can report local and online scams, and investigators or moderators can review those reports to spot recurring fraud patterns. Right now, most people who get scammed either stay quiet or post about it in random social media groups where the warning quickly gets buried. This project aims to give communities one organized place to log scam details—like fake phone numbers, fraudulent bank accounts, suspicious links, or deceptive job offers—so patterns can be identified early and fewer people fall for the same tricks.

2. Key Problems Addressed

- **Scattered Information:** Scam warnings are usually spread across different WhatsApp forwards, Facebook posts, and word-of-mouth conversations, making it hard to check if a number or seller is already known for fraud.
- **Lack of Pattern Tracking:** When the same scammer uses one phone number or bank account to target ten different people, individual victims do not realize it is part of a larger pattern.
- **Slow Investigation and Verification:** Without a structured way to group similar complaints together, moderators and community admins waste time reviewing duplicate or disconnected reports instead of focusing on repeat offenders.
- **Limited Public Awareness:** Ordinary users have no simple way to search a suspicious contact, payment detail, or offer before sending money.

3. System Functionality and Behavior

- **Scam Reporting:** Registered community members can submit a structured report describing what happened, the type of scam (such as fake online shopping, phishing links, impersonation, or bogus job offers), the date, approximate financial loss, and any identifiers used by the scammer (phone numbers, social media handles, email addresses, or bank/wallet numbers).

- **Pattern Grouping and Linking:** When multiple reports share the same phone number, payment account, or website link, the system groups them together and flags them as a recurring pattern so investigators can see how widespread the issue is.
- **Investigation Workflow:** Assigned investigators or moderators can review incoming reports, update their status (such as Pending Review, Verified, Under Investigation, or Dismissed), add investigation notes, and link related cases together.
- **Public Lookup and Alerts:** Regular users can search the system using a phone number, account detail, or scam category to see if previous verified warnings exist before they make a transaction.
- **Basic Trend Summaries:** The system provides clear summaries showing which scam types are happening most frequently, which areas or platforms are targeted most, and which scammer identifiers appear repeatedly.

4. Key Scenarios and Use Cases

- **Multiple Victims Reporting the Same Scammer:** Three different students report an online seller who took advance payment via the same mobile wallet number and never delivered the item. The system links all three reports under one active investigation and highlights the repeated wallet number.
- **Pre-Transaction Safety Check:** Before transferring money to an unknown online page or caller, a user searches the phone number in the system and sees two verified scam reports linked to it, helping them avoid losing money.
- **Evolving Scam Tactics:** A scammer changes their social media page name but keeps using the same bank account or contact number. The system still connects the new reports to the older ones through the matching details.
- **Investigator Case Management:** An investigator reviews a batch of new submissions, marks false or incomplete reports as dismissed, verifies genuine reports with clear evidence, and updates the case status as more victims come forward.
- **Community Trend Monitoring:** A community coordinator checks monthly summaries to see a sudden spike in fake internship or scholarship scams and uses that information to warn people.

5. Project Scope

Included in Scope

- User registration and role-based access for regular community reporters, investigators/moderators, and system administrators.
- Submitting, viewing, updating, and categorizing scam reports with scammer contact or payment details.
- Searching and filtering reports by scam type, location, date, status, or reported identifier (phone number, URL, account number).
- Automatic grouping or flagging of reports that share matching scammer identifiers.
- Investigation status tracking, case notes, and basic summary reports on common scam trends.

Excluded from Scope

- Real-time integration with banks, telecom companies, or law enforcement databases.

- Automated blocking of phone numbers, websites, or bank accounts.
- Financial recovery, payment disputes, or legal action on behalf of victims.
- Advanced machine learning or AI-based natural language analysis of message screenshots.
- Live chat support or emergency hotline services.

6. System Limitations and Boundaries

- It will not act as an official police or cybercrime agency portal; it is a community-driven reporting and analysis system.
- It will not freeze bank accounts, trace real-world GPS locations of scammers, or reverse fraudulent transactions.
- It will not automatically verify whether a user's claim is true without human review by an investigator or moderator.
- It will not support anonymous, unverified mass-reporting without an account, in order to reduce spam and fake accusations.
- It will not provide mobile apps or automated SMS/WhatsApp alert bots in this version.